

Sol

i Char c

0 a

1 +

2 b

3 *

4 (

5 c

6 1

7 d

8 -

9 e

10)

11 ^

12 (

13 f

14 +

15 g

16 *

17 h

18)

19 -

20 i

+ → push to stack (top++)

b → add to result

* → top != -1 & * > + → push

(→ push

c → add to result

1 → 3 < -1 F ⇒ push

d → add to result

- → 1 <= 3 T ⇒ pop

add if i to result

e → add to result

) → pop -, add if to result remove (

^ → 3 <= 2 F ⇒ push

(→ push

added to result

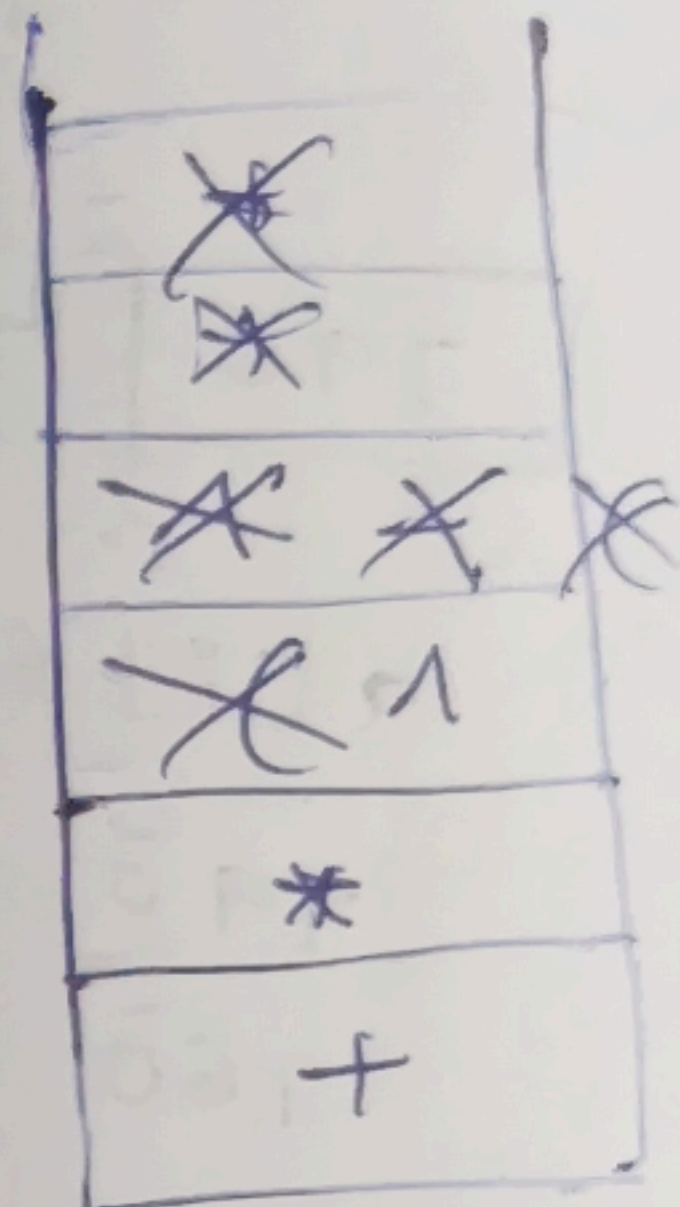
f → ~~push~~ ~~+~~ ~~push~~

+ → 1 <= -1 F ⇒ push

g → add to result

* → * > + ⇒ push

h → add to result



) → pop +
 pop + } Add to result
 remove (

→ 1 ≤ 3T
 pop +

1 ≤ 2T

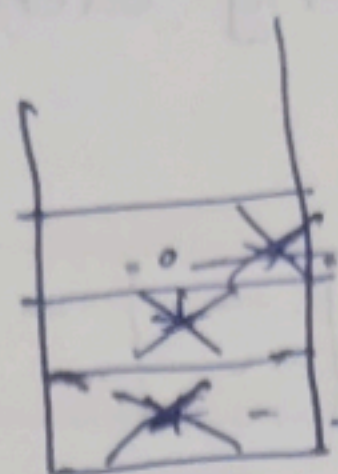
pop +

1 ≤ 1T

pop +

Now, Push -

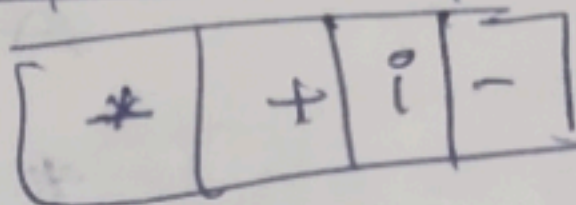
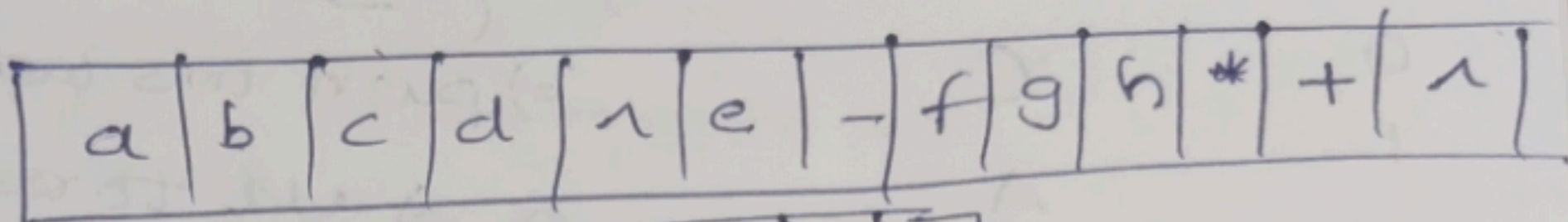
→ add to result



pop (remaining & add to

result

↑
 stack



2) (A * B) - (C - D)

i

char c

c - add to stack

0

(

A → add to result

1

A

* - top! = -1T

2

*

2 ≤ -1F

3

B

⇒ add to stack

4

)

B - add to result

5

-

) - top! = -1T

6

C

stack[top]! = '('T

7

(

add top to result

8

-

← remove '('

9

D

10

)

→ push

C → push to stack

C add to result

→ 1 ≤ 1F

→ push to stack

D → add to result

) → top! = -1

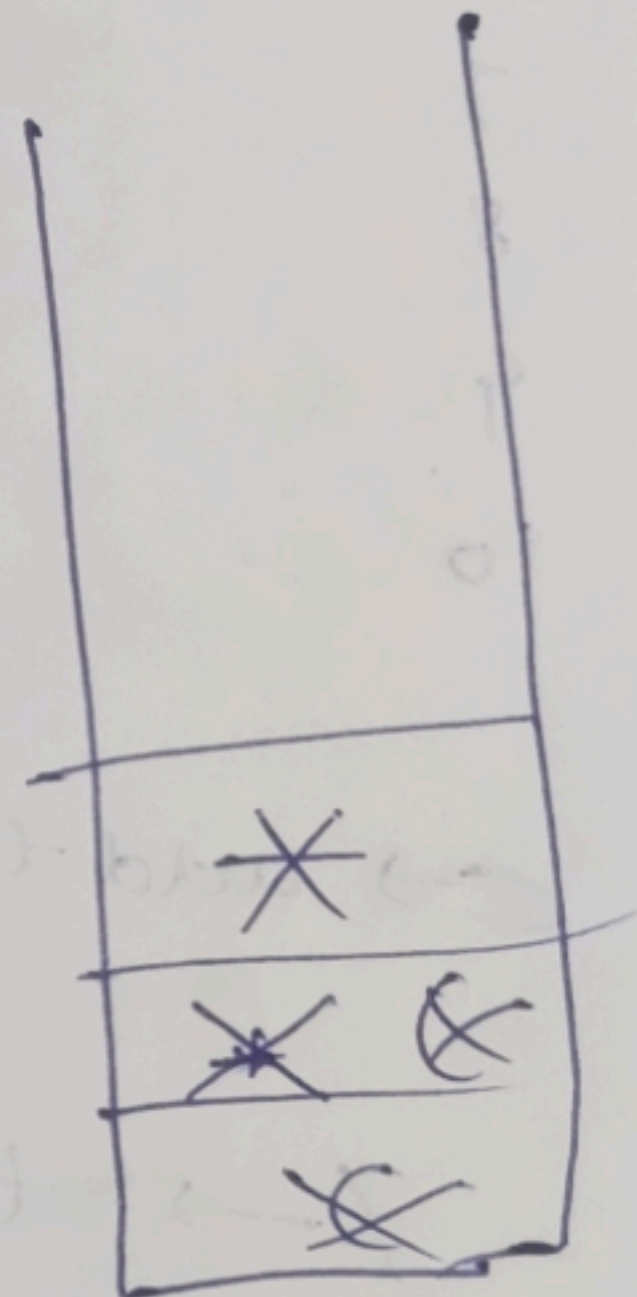
stack[top]! = '('

so, pop - & add

to result

stack[top] = '('

top --

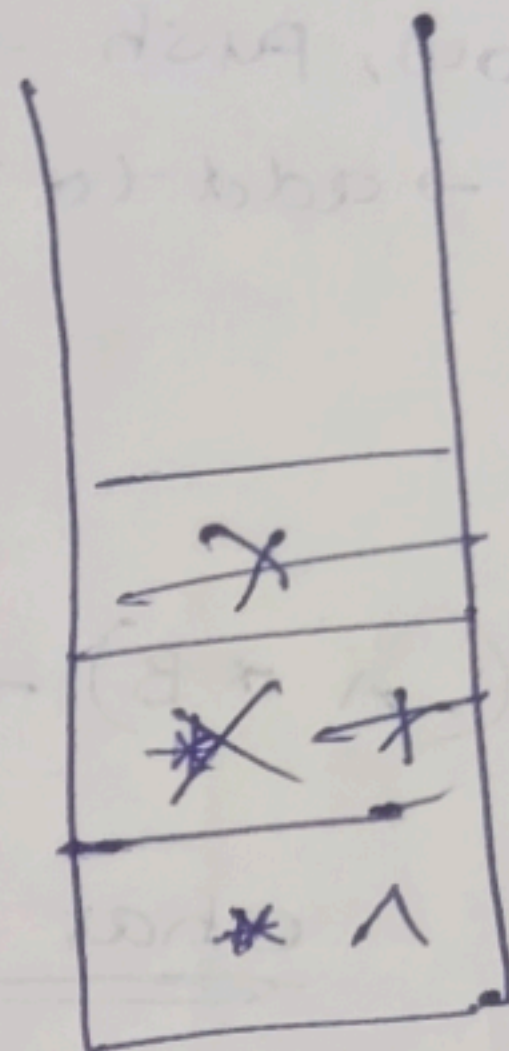


Result :-

A	D	4	C	D	-	-
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9) $(A \times B) \cap (C/D)$

	<u>char-c</u>	
0	(	$\Rightarrow$ push to stack
1	A	$A \Rightarrow$ add to result
2	+	$* \rightarrow \text{top!} = -1 \&$
3	B	$2 < = -1 F$
4	)	$\Rightarrow$ push to stack
5	1	$B \Rightarrow$ add to result
6	(	$') ' \rightarrow \text{top!} = -1 \&$
7	C	$\text{stack} [\text{top}] != 'C' T$
8	/	So, pop $\&$ add to
9	D	result remove '('
10	)	



$\hookrightarrow$ add to result

$C \rightarrow \text{push}$

1. $\rightarrow \text{top!} = -1$, $2 \leq -1 \Rightarrow \text{push}$

D → add to result

$$) \rightarrow \text{top}! = -17$$

stack[top]! = '\0, pop /

remove



Stack

Finally, pop the remaining elements in the stack, i.e. 1.

Final result :-

A	B	*	C	D	/	1
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